Weekly Progress Report

Team #: 41

Project Name: Hospital IoT Simulation Homelab for Cybersecurity Research

Team Members: David Moad

Advisor: Rob Kuschel, Principal Engineer (CTH): GE Aerospace

Date: September 8 – September 14, 2025

What was accomplished last week:

- Finalized preliminary proposal including problem statement, objectives, and methodology
- Completed hardware requirements analysis and budget estimation (~$5,400)
- Researched regulatory compliance requirements (HIPAA, NIST CSF, IEC 62443, ISO/IEC 27001)
- Ordered Dell PowerEdge R740XD server and storage components
- Ordered UniFi networking equipment (UDM-SE, Switch Pro Max 24 PoE, U6 Pro AP)
- Began literature review on hospital cybersecurity threats and IoT vulnerabilities

Plans for coming week:

- Await hardware delivery (expected mid-week)
- Prepare rack space and power infrastructure
- Download TrueNAS CORE ISO and documentation
- Design initial VLAN architecture (IoT, Management, Secure, Guest)
- Order remaining Raspberry Pi 5 kits and SenseHAT components

Issues or concerns:

- Lead time on some hardware components may extend into next week
- Need to finalize rack mounting strategy for optimal cable management

Advisor comment(s): A simple check in with hardware procured. It was critical that we got to brass tax on requirements for the project.

Signature of Advisor/Date: Robert Kuschel

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Team Members: David Moad

Advisor: Rob Kuschel, Principal Engineer (CTH): GE Aerospace

Date: September 15 – September 21, 2025

What was accomplished last week:

- Received Dell PowerEdge R740XD server and Sysracks 18U rack
- Received UniFi UDM-SE gateway and Switch Pro Max 24 PoE
- Physically racked server using Dell 2U B6 sliding readyrails
- Installed CyberPower CP1500 UPS and connected to rack PDU
- Performed initial power-on test of server - verified POST and iDRAC9 functionality
- Connected UniFi equipment for initial configuration
- Received 5x Raspberry Pi 5 kits with PoE+ HATs and SenseHAT displays

Plans for coming week:

- Install TrueNAS CORE on Dell R740XD server
- Configure iDRAC9 for remote management
- Set up initial network configuration with UDM-SE
- Begin VLAN segmentation planning (VLAN 10: IoT, VLAN 20: Medical, VLAN 30: Management, VLAN 40: Guest)
- Configure ZFS storage pools on TrueNAS

Issues or concerns:

- Need to decide on RAID configuration for optimal performance vs redundancy
- iDRAC9 default credentials need to be changed for security

Advisor comment(s): Simple onboarding of hardware and host functionality.

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Team Members: David Moad

Advisor: Rob Kuschel, Principal Engineer (CTH): GE Aerospace

Date: September 22 – September 28, 2025

What was accomplished last week:

• Successfully installed TrueNAS CORE on Dell R740XD using BOSS 2x 240GB M.2 SSDs (RAID 1)
• Configured iDRAC9 with secure credentials and verified remote access capability
• Created ZFS storage pools: RAID 1 on 2x 480GB SSDs (~480GB usable) for VMs
• Set up initial network interface configuration on TrueNAS (X550 10GbE primary, I350 1GbE backup)
• Configured UDM-SE with production-style segmented VLAN architecture:
  * VLAN 10 (10.10.0.0/24): IoT/Patient Sensors
  * VLAN 20 (10.20.0.0/24): Medical Services
  * VLAN 30 (10.30.0.0/24): Secure Management
  * VLAN 40 (10.40.0.0/24): Guest Network
• Connected Switch Pro Max 24 PoE to UDM-SE and verified 10GbE uplink using SFP+ DAC cable

Plans for coming week:

• Deploy first VMs: OPNsense firewall and MQTT broker
• Configure switch port profiles for VLAN assignments
• Begin Raspberry Pi 5 initial setup and OS installation
• Test PoE functionality with Raspberry Pi devices
• Start developing Python scripts for SenseHAT patient vital simulation

Issues or concerns:

• Complex inter-VLAN routing rules need careful planning to avoid misconfigurations
• Need to ensure proper firewall rules between medical services and management networks

• Implementing production-level network segmentation from the start - will monitor complexity

Advisor comment(s): Starting to establish basic backbones of the system.

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Team Members: David Moad

Advisor: Rob Kuschel, Principal Engineer (CTH): GE Aerospace

Date: September 29 – October 5, 2025

What was accomplished last week:

- Deployed OPNsense Firewall VM on TrueNAS (2 vCPU, 4GB RAM, multiple VLAN bridges)
- Configured switch port assignments for VLAN segregation
- Set up ports 2, 3, 4 on switch for VLAN 10 (IoT devices)
- Installed Raspberry Pi OS on all 5 Raspberry Pi 5 units
- Successfully tested PoE+ functionality - Pis powered and networked via single cable
- Attached SenseHAT displays to 3 Raspberry Pis
- Installed required Python libraries (sense-hat, paho-mqtt) on Raspberry Pis
- Configured static IP addressing for Pis on VLAN 10 (10.10.0.20, .21, .22)

Plans for coming week:

- Deploy MQTT broker VM (Mosquitto) for IoT communications
- Develop ECG simulation Python scripts for realistic heartbeat displays
- Create temperature monitoring scripts with encryption (HIPAA compliance)
- Test MQTT publish/subscribe between Raspberry Pis and broker
- Begin documentation of network topology

Issues or concerns:

- Raspberry Pi SenseHAT temperature readings affected by CPU heat - need calibration
- VLAN routing complexity increasing - firewall rules between VLANs proving challenging
- Inter-VLAN communication requiring more troubleshooting than anticipated

Advisor comment(s): Started talking MQTT obligations and HIPAA, HiTech, and Privacy law.

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Advisor: Rob Kuschel, Principal Engineer (CTH): GE Aerospace

Date: October 6 – October 12, 2025

What was accomplished last week:

• Deployed MQTT Broker VM (Ubuntu 24.04, Mosquitto) on TrueNAS - 2 vCPU, 4GB RAM
• Configured Mosquitto with TLS encryption for secure communications
• Developed HeartRate1Monitor.py - ECG-style heartbeat sweep display using Gaussian waveform modeling
• Developed HeartRate2Monitor.py - Secondary ECG simulator with different patient ID
• Successfully tested ECG scripts on Raspberry Pi with SenseHAT - 62 BPM realistic waveform
• Created temp_scroll.py - basic temperature display script
• Developed secure_temp_logger.py - HIPAA-compliant encrypted temperature logging using Fernet
• Generated encryption key and stored securely: /home/r-pi-1/.keys/patient_zero.key

Plans for coming week:

• Deploy Node-RED VM for IoT workflow automation
• Test end-to-end MQTT data flow: Pi sensors → MQTT broker → Node-RED
• Configure systemd services for persistent ECG monitoring across reboots
• Begin PostgreSQL database VM deployment for data storage
• Create initial Grafana dashboard mockup for vital signs visualization

Issues or concerns:

• Attempted cloud storage integration (OneDrive/rclone) - proved problematic and unreliable
• Decided to use local file storage instead for better reliability
• VLAN segmentation causing significant connectivity issues between IoT devices and VMs
• Need to document encryption key management procedures for compliance

Advisor comment(s): Talked about how layered approach for defense in depth can serve as hurdle towards development and what the primary focus should be on and at what phase. Security is paramount early in the project, but cannot hinder dev work. This is especially critical while the system has no production data and security is not necessary for data protection.

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Advisor: Rob Kuschel, Principal Engineer (CTH): GE Aerospace

Date: October 13 – October 19, 2025

What was accomplished last week:

• Deployed Node-RED VM (Ubuntu, 2 vCPU, 4GB RAM) on VLAN 20
• Configured Node-RED flows for MQTT data ingestion and processing
• Attempted MQTT publish from Raspberry Pi to broker - encountering inter-VLAN routing issues
• Configured systemd service files for HeartRate1Monitor.py and HeartRate2Monitor.py
• Verified ECG monitoring scripts persist across system reboots
• Set up ID1.py script for patient identification display with blinking effect
• Tested encrypted temperature logging - verified 30-second intervals and Fernet encryption
• Successfully decrypted temperature logs using decrypt_log.py

Plans for coming week:

• Troubleshoot and resolve VLAN connectivity issues
• Integrate MQTT data flow with Node-RED processing
• Begin PostgreSQL VM deployment
• Create database schema for storing patient vital signs
• Update Python scripts to publish sensor data to MQTT topics

Issues or concerns:

• CRITICAL: Multi-VLAN architecture creating severe complexity in data flows
• Firewall rules between VLANs requiring extensive troubleshooting - preventing proper IoT integration
• Raspberry Pis on VLAN 10 cannot reliably communicate with MQTT broker on VLAN 20
• LESSON LEARNED: Jumped directly to production-type segmented network architecture too early

• May need to simplify network architecture to prioritize functionality before security segmentation

Advisor comment(s): Started talking about simplicity approaches. Two paths 1:1 NAT or flat network for development.

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Advisor: Rob Kuschel, Principal Engineer (CTH): GE Aerospace

Date: October 20 – October 26, 2025

What was accomplished last week:

• Attempted MQTT broker and Node-RED integration - unsuccessful due to VLAN issues
• Verified TLS 1.3 encryption configuration on MQTT (when accessible)
• Documented current VM infrastructure: mqtt_broker_vm and node_red_vm deployed but not fully functional
• Created network topology diagram showing complex VLAN segmentation
• Extensively tested inter-VLAN routing rules via OPNsense firewall - persistent connectivity problems
• Researched best practices for phased network deployment in lab environments
• Made critical project decision: flatten network architecture temporarily
• Backed up TrueNAS configuration to laptop for disaster recovery before making changes

Plans for coming week:

• Redesign network architecture - flatten VLANs to single subnet (VLAN30: 10.30.0.0/24) for initial build
• Create NETWORK_SETUP_PROGRESS.md document to track configuration changes and lessons learned
• Execute migration from segmented multi-VLAN to flat single-subnet architecture
• Prepare for network reconfiguration while preserving VM functionality
• Document decision rationale and plan for future VLAN reinstatement

Issues or concerns:

• MAJOR LESSON LEARNED: Attempting production-level security architecture at project start was a mistake
• Multi-VLAN segmentation preventing completion of core IoT functionality
• Raspberry Pis losing connectivity, data flows blocked by complex firewall rules

• DECISION: Prioritize building functional infrastructure first, add security layers incrementally later
• Will reinstate multi-level segmented VLANs once core data pipeline is stable and tested
• This reflects real-world principle: establish baseline functionality before implementing complex security

Advisor comment(s): We discussed getting systems working before incorporating network security elements. This tends to be very challenging. And while critical in production, it can be very painful in non-production or develipment environments. We'll likely re-inforce with a vulnerability scan when complete to determine reach from different parts of the network segment.

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Advisor: Rob Kuschel, Principal Engineer (CTH): GE Aerospace

Date: October 27 – November 2, 2025

What was accomplished last week:

• Created comprehensive NETWORK_SETUP_PROGRESS.md documenting architectural pivot
• Backed up TrueNAS configuration (restoration file saved to laptop)
• Successfully flattened network to single subnet (VLAN30: 10.30.0.0/24):
  * UDM-SE: 10.30.0.1 (gateway)
  * TrueNAS: 10.30.0.44/24 on br0 interface
  * DHCP range: 10.30.0.6 - 10.30.0.254
• Removed test IP configurations from TrueNAS (cleaned up 192.168.1.10)
• Reconfigured switch ports 2, 3, 4 from VLAN10 to VLAN30 - all devices now on same subnet
• Power cycled Raspberry Pis via PoE to force DHCP renewal on new network
• Verified 2 Pis still running ECG scripts (HeartRate1Monitor and HeartRate2Monitor)
• Confirmed mqtt_broker_vm and node_red_vm operational after network simplification

Plans for coming week:

• Manually check each Raspberry Pi to find new IP addresses (10.30.0.x range)
• SSH to each Pi and verify connectivity to TrueNAS (10.30.0.44) and MQTT broker
• Configure static IP addresses for Pis (recommend 10.30.0.20, .21, .22, .23, .24)
• Test ping between all devices on new flat network - expect immediate success
• Delete OPNsense VM from TrueNAS (no longer needed with simplified architecture)
• Document plan for eventual VLAN reinstatement after core functionality proven

Issues or concerns:

• Raspberry Pis may have cached old network configurations - need manual verification
• Unknown if all Pis successfully obtained new DHCP addresses on 10.30.0.x subnet

• Need systematic approach to discover Pi IPs without causing disruption
• NOTE: This flat network is TEMPORARY - will reinstate security segmentation in future phase

Advisor comment(s): The flat nework approach is paramount, as David and I discussed seldom is it easy to develop in a secured state and this is the critical path for having a development environment that you harden into production through a QA or acceptance environment. He's on the right path with this adjustment path.

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Team Members: David Moad

Advisor: Rob Kuschel, Principal Engineer (CTH): GE Aerospace

Date: November 3 – November 9, 2025

What was accomplished last week:

• Successfully located all Raspberry Pi IP addresses on new 10.30.0.0/24 network
• Verified connectivity via SSH to operational Raspberry Pis - IMMEDIATE SUCCESS
• Confirmed HeartRate1Monitor.py and HeartRate2Monitor.py scripts still running perfectly
• Tested network connectivity: Pis can now reach TrueNAS (10.30.0.44) and UDM-SE (10.30.0.1)
• BREAKTHROUGH: MQTT broker VM now accessible from Raspberry Pis - no firewall blocking
• Deleted OPNsense VM from TrueNAS - unnecessary with flat network architecture
• Updated network documentation with current simplified topology
• Tested "portable mode" - disconnected WAN cable, verified lab functions standalone
• Network simplification vindicated - all connectivity issues resolved

Plans for coming week:

• Configure static IP addresses for all Raspberry Pis (10.30.0.20-24)
• Update Raspberry Pi scripts to publish data to MQTT broker on 10.30.0.51
• Deploy PostgreSQL database VM (10.30.0.50 recommended)
• Deploy Grafana/Prometheus VM (10.30.0.52 recommended)
• Create MQTT → PostgreSQL data pipeline (now possible with flat network)
• Begin building Grafana dashboards for real-time vital signs visualization

Issues or concerns:

• Need to update all Python scripts with new MQTT broker IP address
• Static IP assignment will require editing /etc/dhcpcd.conf on each Pi
• Should document IP allocation scheme for future VM deployments (10.30.0.50+)

• Must plan timeline for reinstating multi-level VLAN segmentation once core infrastructure proven

Advisor comment(s): We discussed a variety of MQTT brokers and the use of it as a common industrial protocol. ZeroMQTT was one such platform we discussed. We also discussed the need for using static IP Addresses in OT enforced with static DHCP reservation to keep other devices from grabbing the same IP Address.

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Team Members: David Moad

Advisor: Rob Kuschel, Principal Engineer (CTH): GE Aerospace

Date: November 10 – November 16, 2025

What was accomplished last week:

• Assigned static IP addresses to operational Raspberry Pis (10.30.0.20, .21)
• Updated MQTT broker configuration to accept connections on 10.30.0.51
• Successfully verified end-to-end connectivity: Raspberry Pis → MQTT Broker → Node-RED
• Documented IP allocation scheme:
  * 10.30.0.1: UDM-SE Gateway
  * 10.30.0.44: TrueNAS
  * 10.30.0.20-24: Raspberry Pi IoT sensors
  * 10.30.0.50+: Future VM infrastructure (PostgreSQL, Grafana, FHIR, PACS, Wazuh, etc.)
• Created VM deployment priority list organized in four tiers
• Prepared 5-minute capstone presentation demonstrating project scope and current progress
• Documented encryption key: 732a7722df86ee181382dfc13a5fb02041a09f341174283a9e0d4f4ef79ae9c4
• Reflected on architectural decision: Lesson learned about phased implementation approach

Plans for coming week:

• Deploy PostgreSQL database VM for storing patient vital signs
• Configure database schema for ECG data, temperature logs, and timestamps
• Update Python scripts to publish structured JSON data via MQTT
• Create Node-RED flow to insert MQTT data into PostgreSQL
• Begin Grafana VM deployment for dashboard visualization
• Prepare demonstration of current ECG monitoring functionality
• Plan future phase: Reinstate multi-level segmented VLANs after core data pipeline proven stable

Issues or concerns:

- Need to decide on database schema design for optimal query performance
- Consider time-series database (TimescaleDB) vs traditional PostgreSQL
- Remaining 3 Raspberry Pis need configuration and script deployment
- AI camera Pi is lower priority - focus on vital signs monitoring first
- Future work: Document plan for phased VLAN segmentation reinstatement
- Key takeaway: Building complex security architecture requires functional foundation first

Advisor comment(s): Solid work on MQTT broker and looking at time series DB, made a mild recommendation to consider Apache IOTDB asa n open source option.

We also disucssed thoughts around simulated Red Team attacks.

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